

International Marketing Analytics

Open-source marketing mix modelling

WIRTSCHAFTS UNIVERSITÄT WIEN VIENNA UNIVERSITY OF ECONOMICS AND BUSINESS

Dr Arne Floh, Senior Lecturer in Marketing
WU Vienna, Institute for International Business

arne.floh@wu.ac.at
Office hours: book online (Google Calendar)

WT 2026/27

Today

Concepts → By hand → Three tools → Real data → Python → In class

Theory

- The model in one line, carryover, saturation, ROI
- A small example by hand
- PyMC-Marketing, Meridian, Robyn

Practice

- Real data: a sports web shop in Austria
- Python: a code ladder in nine steps
- Online quiz and an in-class assignment (no coding)

Question of the day: where should the next euro go, and how sure can we be?

Our example: where should the next euro go?

The problem: a brand spends on TV and online and must split next year's budget.

The question: what did each channel bring, and where should the next euro go?

Measure

sales from each channel

Explain

baseline vs media

Plan

move budget to the best next euro

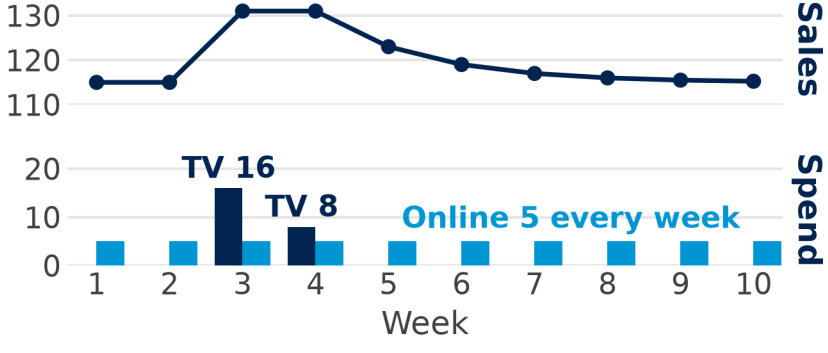
A tiny coffee brand first: 10 weeks, TV and online, by hand; a real web shop at the end.

Invented round numbers (hand_example.csv)

The data: ten weeks, one row per week

Coffee brand: spend and sales per week, k EUR

Week	1	2	3	4	5	6	7	8	9	10
TV	0	0	16	8	0	0	0	0	0	0
Online	5	5	5	5	5	5	5	5	5	5
Sales	115	115	131	131	123	119	117	116	115.5	115.25



Sales jump in the TV weeks 3 and 4: 131

and fade slowly: 123, 119, 117 …

Online: 5 k EUR every week

Invented round numbers (hand_example.csv)

The main concepts

WIRTSCHAFTS UNIVERSITÄT WIEN VIENNA UNIVERSITY OF ECONOMICS AND BUSINESS

The model in one line, carryover, saturation, ROI

Why MMM is back

Tracking breaks

Apple's opt-in rule and the GDPR hide many user journeys from the ad platforms.

No user data

Only weekly totals of sales and spend per market.

Free tools

PyMC-Marketing, Meridian and Robyn are open source.

Old idea, new tools: MMM dates back to the 1960s; Bayesian methods and free code made it affordable.

MMM works where tracking fails: it needs only weekly totals per market.

The model in one line

Sales = baseline + media effects + controls + noise

Baseline

what you would sell with no media: trend, season, holidays

Media effects

each channel after carryover and saturation

Controls

price, promotions, competitors, weather

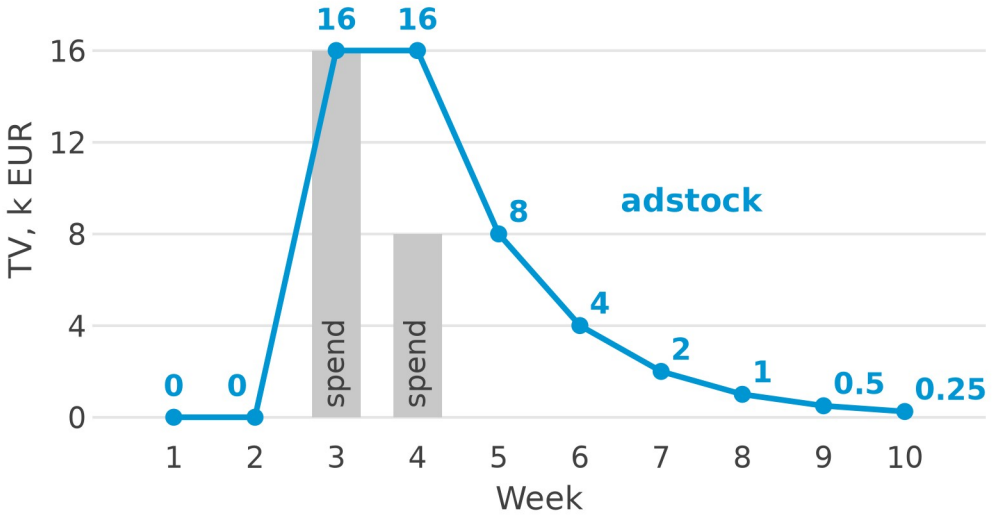
Coffee brand, week 3: 100 + 1 × 16 + 3 × 5 = 131

baseline 100 · TV adstock × 1 · online × 3

The output: the contribution and ROI of each channel, response curves, a budget plan.

Invented round numbers (hand_example.csv)

Carryover: this week's spend keeps working



Adstock = this week's spend + **decay** × last week's adstock

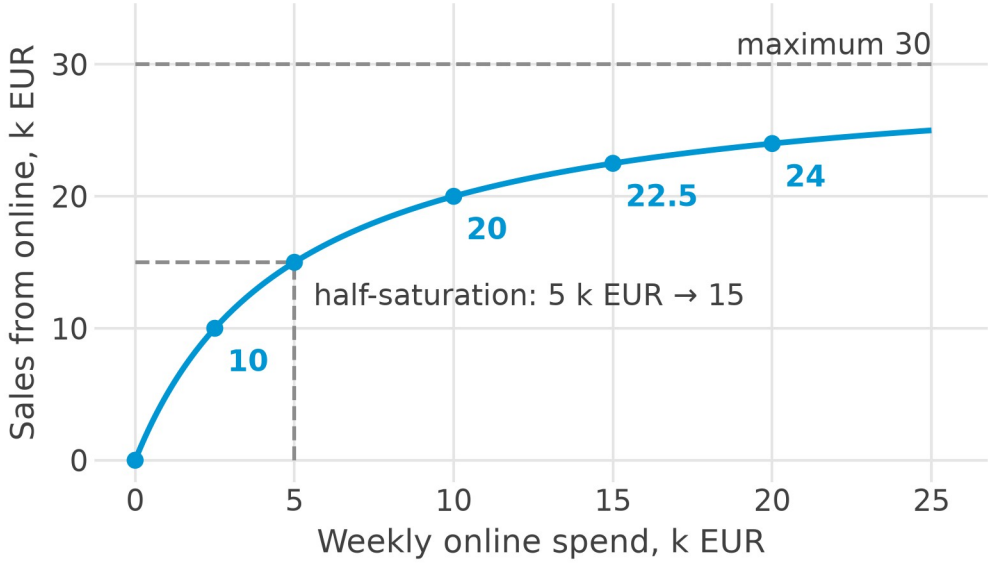
0.5 **decay:** half of last week's adstock is left this week

× 2 **each euro works twice:** 1 + 0.5 + 0.25 + 0.125 + … = 2

Carryover: TV in weeks 3–4 still sells in weeks 5–10.

Invented round numbers (hand_example.csv)

Saturation: each extra euro buys less



Online effect =
30 × spend / (spend + 5)

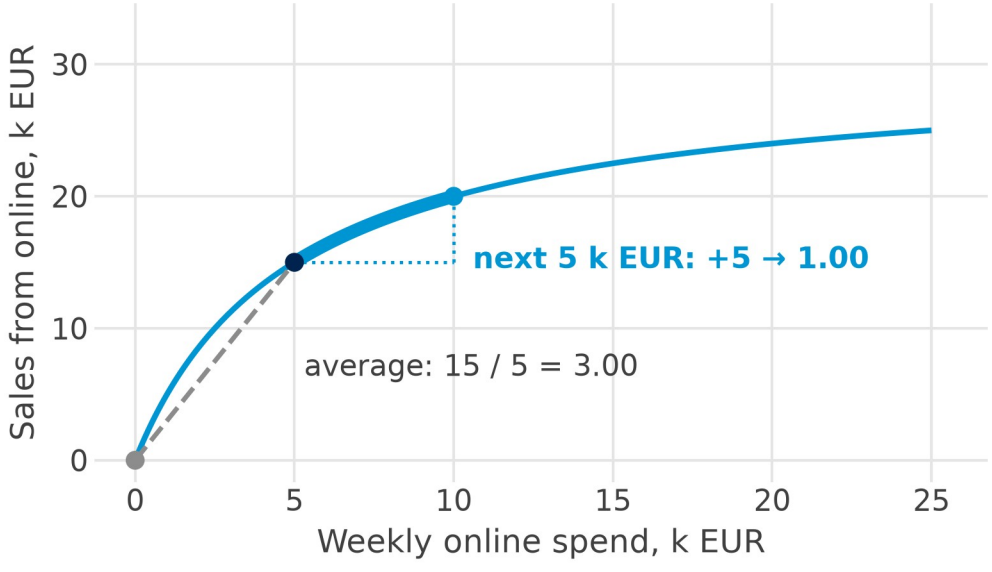
30 **maximum:** the most online can bring per week

5 **half-saturation:** at 5 k EUR online brings half of it, 15

Saturation: 10 k EUR bring 20, not twice 15.

Invented round numbers (hand_example.csv)

ROI and the response curve



3.00 **average ROI:** 15 / 5: 3 EUR of sales per euro

1.00 **next 5 k EUR:** (20 − 15) / 5 = 1.00

Response curve: weekly sales from a channel against its weekly spend

The shape of the curve, not the average ROI, decides where the next euro goes.

Invented round numbers (hand_example.csv)

A small example by hand

Adstock, contributions, ROI, a budget question

Adstock week by week

Week	1	2	3	4	5	6	7	8	9	10
TV spend	0	0	16	**8**	0	0	0	0	0	0
+ 0.5 × last	0	0	0	**8**	8	4	2	1	0.5	0.25
= adstock	**0**	**0**	**16**	**16**	**8**	**4**	**2**	**1**	**0.5**	**0.25**

Week 4: 8 + 0.5 × 16 = 16
new spend + half of last week's adstock

24
TV spend
in k EUR

Week 5: 0 + 0.5 × 16 = 8
no new spend: half is left

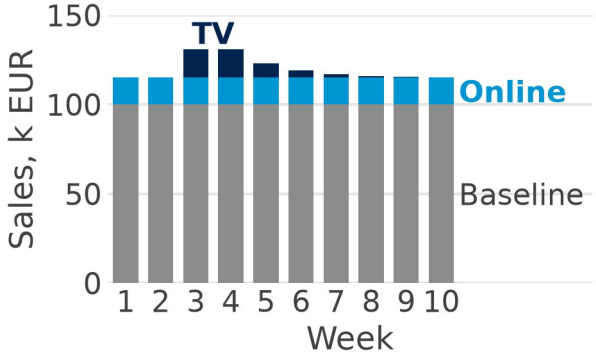
47.75
TV adstock
weeks 1–10

Each euro works twice: 24 k EUR of TV spend count as 47.75 in ten weeks, 48 in the long run.

Invented round numbers (hand_example.csv)

Contributions and ROI by hand

Part	Calculation	Sales	Spend	ROI
Baseline	100 × 10 weeks	1,000	–	–
TV	1 × 47.75 adstock	47.75	24	1.99
Online	3 × 50 spend	150	50	3.00
Total		**1,197.75**	**74**	–



Sales per week: baseline, online, TV

ROI = sales ÷ spend. **Shares** of 1,197.75: baseline 83.5 %, online 12.5 %, TV 4.0 %

Ignore carryover and TV looks half as good: 24 / 24 = 1.00 instead of 1.99.

Invented round numbers (hand_example.csv)

The budget question: 5 k EUR more a week

The question: the brand has 5 k EUR more a week. TV or online?

TV

5 k EUR more TV work twice (carryover):

1 × 5 × 2 = 10 k EUR of sales

ROI of the next euro: 10 / 5 = 2.00

Online

Its curve (slide 9): 15 at 5, 20 at 10 k EUR:

20 − 15 = 5 k EUR of sales

ROI of the next euro: 5 / 5 = 1.00

Average ROI: online 3.00, TV 1.99: the average would pick the wrong channel.

Decide at the margin: put the next euro where the next euro sells most.

Invented round numbers (hand_example.csv)

Three open-source tools

PyMC-Marketing, Meridian, Robyn

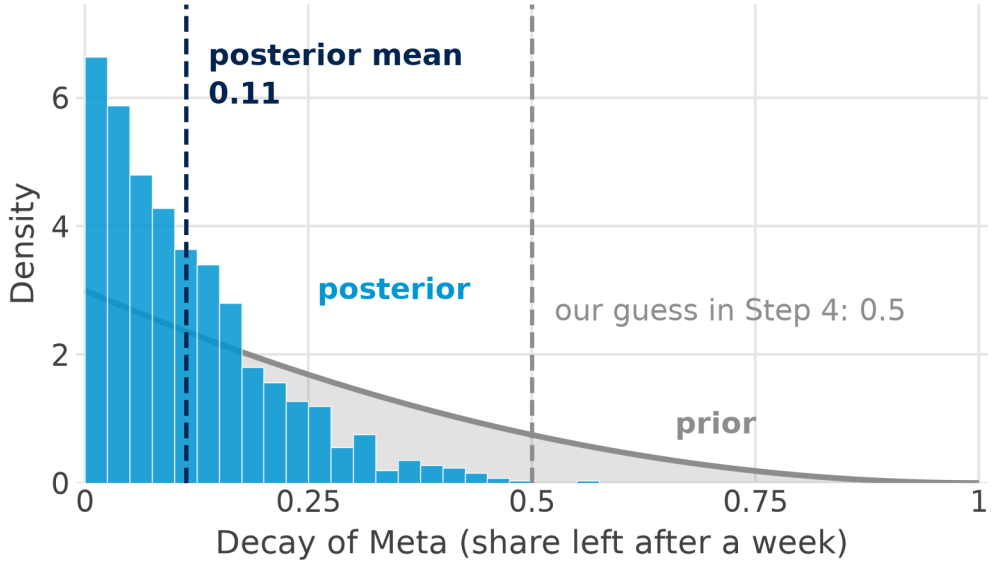
The three tools at a glance

	PyMC-Marketing	Google Meridian	Meta Robyn
Maintainer	PyMC Labs	Google	Meta Marketing Science
Version (Oct 2026)	1.3.0	2.1.0	3.12.1 (CRAN)
Language	Python	Python	R (Python port in beta)
Approach	Bayesian	Bayesian, geo-level	Ridge regression + search
Engine	PyMC	JAX	glmnet, Nevergrad, Prophet
Licence	Apache-2.0	Apache-2.0	MIT

All free and open, but two come from the largest ad platforms (slide 24).

Sources: PyPI, CRAN and the tools' repositories, checked 26 Oct 2026

Bayesian in one picture: prior and posterior



Prior: what we believe before the data (here: the decay is probably small, Beta(1, 3))

Posterior: the belief after the data: mean 0.11, 94 % interval 0.00 to 0.29

Check: R-hat 1.00, no divergences: the sampler agrees with itself

Bayesian MMM: every parameter gets a prior; the result is a range, not one number.

Prior: PyMC-Marketing default; posterior: the Austrian model of Step 8

PyMC-Marketing: Bayesian MMM in Python

What: PyMC Labs' free Python package (Apache-2.0), built on PyMC

How: a prior on every parameter; adstock, then saturation; every result comes with a range

Strengths: flexible; countries and brands in one model; lift tests and a budget optimiser built in

Best for: many markets or brands, a budget split across them, a team that codes in Python

open_source_mmm.qmd · Step 8

```
mmm = MMM(date_column="week", target_column="sales",
          channel_columns=["google", "meta"],
          control_columns=["week_no"],
          adstock=GeometricAdstock(l_max=6),
          saturation=LogisticSaturation(),
          yearly_seasonality=2)
```

After the fit you get	Slide
a posterior for every parameter	17, 39
contributions and ROI with a range	28
response curves, a budget plan	29

Our tool in the code ladder: six lines define the model, one call fits it (Step 8).

Meridian: Google's Bayesian geo model

What: Google's free Python package (Apache-2.0); runs on JAX, a GPU helps

How: priors, as in PyMC-Marketing; regions (geos) in one model, each around a shared mean

Strengths: lift tests become ROI priors; reach and frequency for video; a guided Colab notebook

Best for: data by region and Google-heavy media

Not run here: not in the course environment

```
from meridian.data import (
    data_frame_input_data_builder as db)
from meridian.model import model, spec
builder = db.DataFrameInputDataBuilder(
    kpi_type="non_revenue",
    default_kpi_column="conversions")
builder = (builder.with_kpi(df)
           .with_population(df)
           .with_media(df,
               media_cols=["Channel0_impression"],
               media_spend_cols=["Channel0_spend"],
               media_channels=["Channel0"]))
mmm = model.Meridian(input_data=builder.build(),
                     model_spec=spec.ModelSpec())
mmm.sample_posterior(n_chains=10, n_adapt=1000,
                     n_burnin=500, n_keep=1000)
```

Geo first: Meridian needs population per region and models sales per head.

Source: Meridian Getting Started notebook, github.com/google/meridian (commit ef585b5); not run here

Robyn: Meta's ridge regression and search

What: Meta's free R package (MIT); a Python port is in beta

How: ridge regression (it shrinks the effects to keep them stable); a search tries thousands of adstock and saturation settings

Strengths: a quick first read-out; holidays for 123 countries built in; no sampler to check

Pick: the Pareto front keeps the models no other beats on fit and plausibility; the analyst picks one

Not run here: Robyn is an R package

```
library(Robyn)
InputCollect <- robyn_inputs(
  dt_input = dt_simulated_weekly,
  dt_holidays = dt_prophet_holidays,
  date_var = "DATE", dep_var = "revenue",
  dep_var_type = "revenue",
  prophet_vars = c("trend", "season", "holiday"),
  prophet_country = "DE",
  paid_media_spends = c("tv_S", "search_S"),
  adstock = "geometric",
  hyperparameters = hyperparameters)
OutputModels <- robyn_run(
  InputCollect = InputCollect,
  iterations = 2000, trials = 5)
```

Trade-off: priors make assumptions explicit; the Pareto front makes the analyst's choice explicit.

Source: Robyn demo.R, github.com/facebookexperimental/Robyn (commit aab4226); not run here

Calibrating with experiments

Lift test

Switch a channel on or off in some regions and measure the difference.

Example: more Meta in half of the regions, none in the rest.

Calibration: use that difference to anchor the model.

→ **Meridian:** the lift test becomes the ROI prior of that channel

→ **PyMC-Marketing:** each test is a point on the response curve, per country if needed

→ **Robyn:** tests are an extra objective; models far from the lift lose

International angle: tests often run only in large markets; a pooled Bayesian model passes that evidence on to the small ones.

Many countries: three ways to pool

No pooling



one model per country; small markets get noisy estimates

Robyn

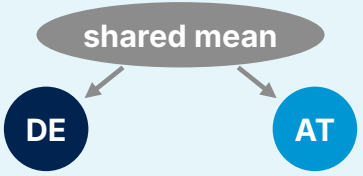
Full pooling



one effect for all countries; hides real differences

rarely sensible

Partial pooling



country effects around a shared mean; small markets borrow strength

Meridian, PyMC-Marketing

Our brand: Austria sold 3,180 k EUR in the same 134 weeks in which Germany sold 11,270

One currency first: Switzerland is in CHF; convert before you pool (ECB rates).

Data: Conjura MMM dataset (Anderson 2024), CC BY 4.0

Which tool for which situation?

PyMC-Marketing

- many countries or brands in one model
- a budget split across markets
- a team that codes in Python

Meridian

- geo-level data and Google-heavy media
- reach and frequency for video
- a guided Colab workflow

Robyn

- one market, few channels, R users
- a quick first read-out
- country holidays built in

Our shop: 2.6 years of weekly data in Austria; Germany next door

Rule of thumb: two or more years of weekly data per market; with less, pool countries or use informative priors.

Reading vendor claims: four questions

1. **Who wrote it?** Google, Meta or a partner may favour their own channels.
2. **Compared with what?** An uplift needs a baseline and a time window.
3. **Tested how?** A holdout period or a lift test, not only in-sample fit.
4. **Which version?** Results on old releases may no longer hold.

Free software is not a free model: the platforms that sell the media also wrote two of the tools that judge it.

Real data: a sports web shop

The data, a warning, the model, the next euro

The data: 134 weeks of a real web shop

File: sports_weekly.csv, Austria (AT)

week	sales	google	meta
26 Jul 2021	7.46	1.02	0.39
2 Aug 2021	9.32	1.36	0.27
9 Aug 2021	8.61	1.28	0.26
… 134 weeks in all			

sales: goods ordered in the web shop, k EUR

google: spend on all Google ads, k EUR

meta: spend on Facebook and Instagram ads, k EUR

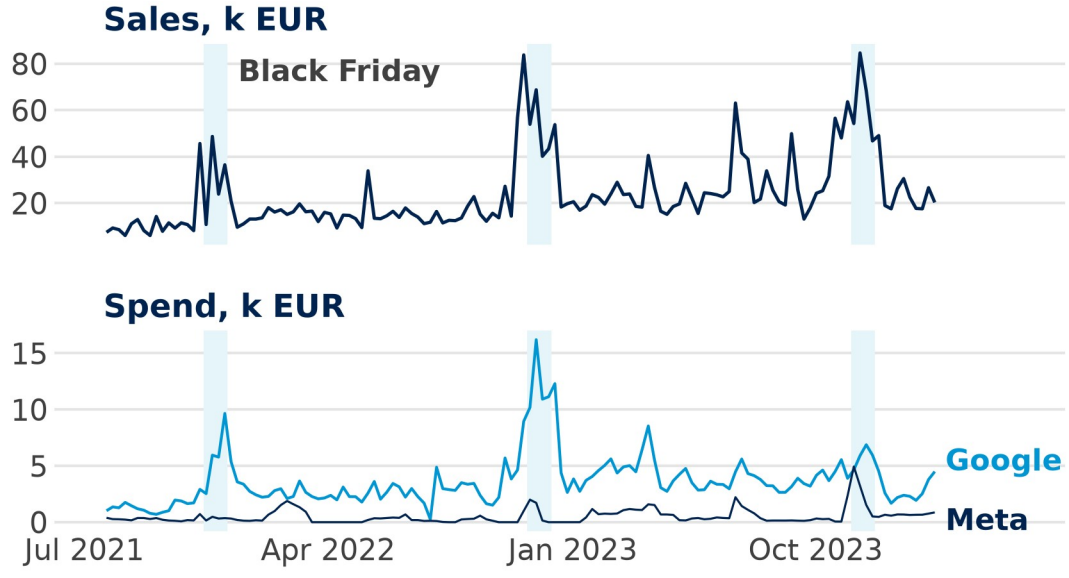
3,180 **k EUR of sales** in 134 weeks

561 **k EUR of spend:** Google 87.5 %, Meta 12.5 %

2.6 **years:** Germany and Switzerland are in the file too

Data: Anderson (2024), Conjura multi-region MMM dataset, figshare, CC BY 4.0; weekly sums by the course

Spend follows demand: a warning



7.03 **k EUR on Google** a week in the 10 best weeks ...

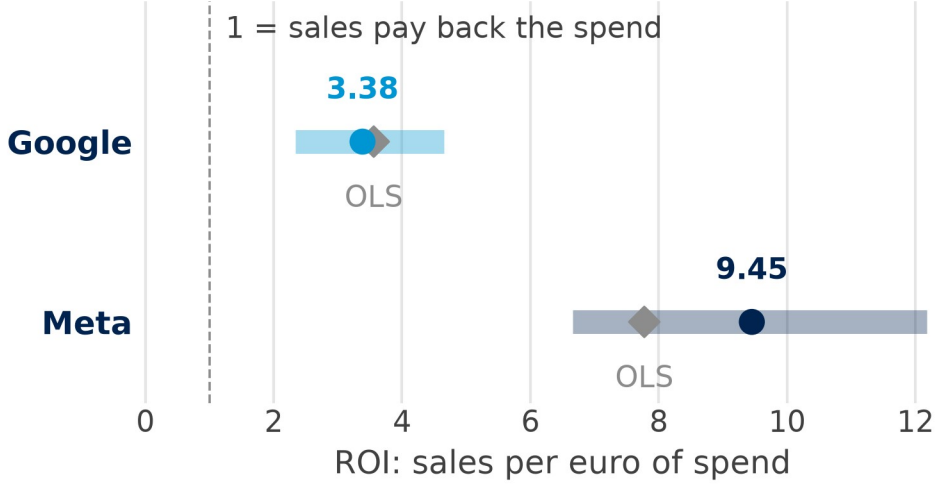
3.39 **... and** in the other 124 weeks

0.68 **correlation** of sales and Google spend

Spend follows demand: shopping ads spend more when more people search.

Data: Conjura MMM dataset (Anderson 2024), CC BY 4.0

What the model says: contributions and ROI



ROI with 94 % interval (PyMC-Marketing); ◆ OLS of Step 6

52 % **of sales from Google** (1,660 k EUR)

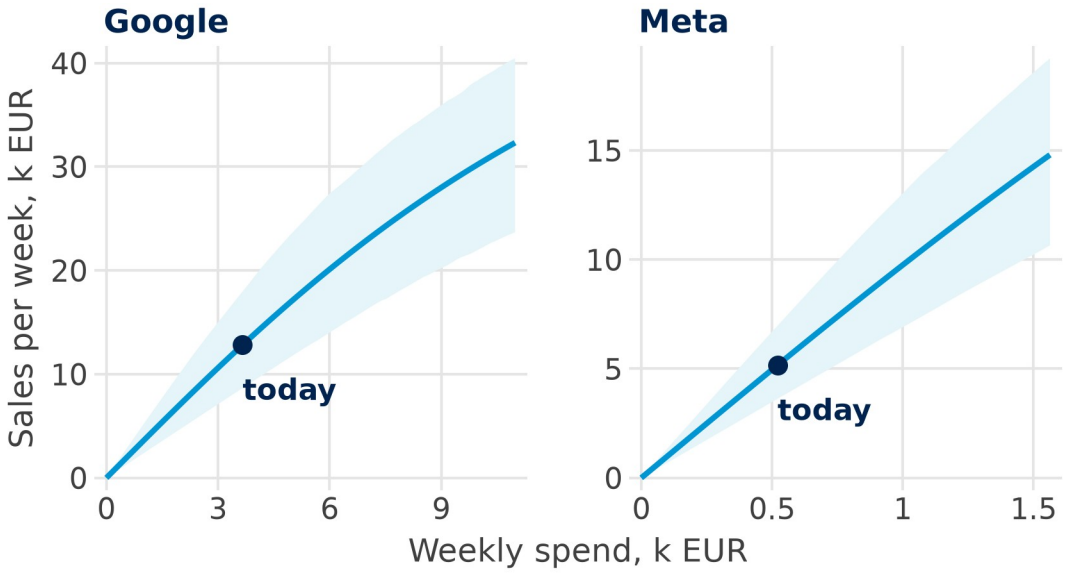
21 % **from Meta** (660 k EUR)

27 % **baseline,** trend and season

Meta: 9.45 per euro (6.66 to 12.18); Google: 3.38 (2.33 to 4.65).

Data: Conjura MMM dataset (Anderson 2024), CC BY 4.0; model: Step 8

Response curves: where the next euro works



9.77 **Meta:** next-euro ROI at today's 0.52 k EUR a week

3.28 **Google:** next-euro ROI at 3.66 k EUR a week

+3.1 **k EUR of sales a week** if 0.5 k EUR move from Google to Meta

Shift a little, then test: a geo lift test in Austria before the big move.

Data: Conjura MMM dataset (Anderson 2024), CC BY 4.0; curves: posterior of Step 8

MMM in Python



A code ladder, one step at a time

Step 1: load and look

open_source_mmm.qmd · Step 1

```
import polars as pl

# load the weekly data and keep Austria
file = "../../data/mmm_open/sports_weekly.csv"
df = pl.read_csv(file, try_parse_dates=True)
at = df.filter(pl.col("country") == "AT")
print(at.head())

# total sales and spend, k EUR
at.select(pl.col("sales", "google", "meta").sum())
```

week	sales	google	meta
26 Jul 2021	7.46	1.02	0.39
2 Aug 2021	9.32	1.36	0.27
9 Aug 2021	8.61	1.28	0.26
total	3,180	491	70

Look first: 134 weeks; Google gets 7 of every 8 euros.

Step 2: a first regression

open_source_mmm.qmd · Step 2

```
import statsmodels.formula.api as smf

# sales on the spend of both channels
m1 = smf.ols("sales ~ google + meta",
             data=at.to_pandas()).fit()
m1.params
```

term	coef
Intercept	5.14
google	3.93
meta	8.10
R^2	0.58



Before any control: a k EUR on Meta goes with 8.10 k EUR of sales.

Step 3: add the trend

open_source_mmm.qmd · Step 3

```
# week_no counts the weeks 1 to 134: the trend
at = at.with_columns(
    week_no=pl.int_range(1, pl.len() + 1))
m2 = smf.ols("sales ~ google + meta + week_no",
             data=at.to_pandas()).fit()
m2.params
```

term	coef
Intercept	0.57
google	3.60
meta	6.93
week_no	0.09
R^2	0.62

The trend takes its share: sales grew without the channels too.

Step 4: adstock in one line

open_source_mmm.qmd · Step 4

```
from scipy.signal import lfilter

# adstock: this week's spend + 0.5 x last stock
at = at.with_columns(
    meta_ad=lfilter([1], [1, -0.5], at["meta"]))
at.select("week", "meta", "meta_ad").head()
```

week	meta	meta_ad
26 Jul 2021	0.389	0.3890
2 Aug 2021	0.273	0.4675
9 Aug 2021	0.259	0.4928
16 Aug 2021	0.231	0.4774
23 Aug 2021	0.167	0.4057

As by hand: 0.273 + 0.5 × 0.389 = 0.4675 (slide 12).

Step 5: regression with carryover

open_source_mmm.qmd · Step 5

```
# Meta with its carryover, Google without
m3 = smf.ols("sales ~ google + meta_ad + week_no",
             data=at.to_pandas()).fit()
m3.params
```

term	coef
Intercept	0.58
google	3.56
meta_ad	3.93
week_no	0.09
R^2	0.60

Per k EUR of adstock: each euro of Meta counts about twice (Step 6).

Step 6: contributions and ROI

open_source_mmm.qmd · Step 6

```
# contribution = coefficient x column total
b = m3.params
c_google = b["google"] * at["google"].sum()
c_meta = b["meta_ad"] * at["meta_ad"].sum()
print(c_google, c_meta)

# ROI = contribution / spend
print(c_google / at["google"].sum())
print(c_meta / at["meta"].sum())
```

channel	contrib.	ROI
google	1,746	3.56
meta	543	7.78
baseline	891	–



Meta returns more per euro, Google brings more sales.

Step 7: saturation in one line

open_source_mmm.qmd · Step 7

```
# saturation: half-saturation at 20 k EUR a week
at = at.with_columns(
    google_sat=at["google"] / (at["google"] + 20))
m4 = smf.ols(
    "sales ~ google_sat + meta_ad + week_no",
    data=at.to_pandas()).fit()
print(m3.rsquared, m4.rsquared)

# the Google ROI of the curved model
b4 = m4.params
c_sat = b4["google_sat"] * at["google_sat"].sum()
print(c_sat / at["google"].sum())
```

model	R^2	Google ROI
straight	0.60	3.56
curved	0.60	4.79

Same fit, other answer: the data cannot choose the curve.

Step 8: PyMC-Marketing

open_source_mmm.qmd · Step 8

```
from pymc_marketing.mmm import (
    GeometricAdstock, LogisticSaturation)
from pymc_marketing.mmm.multidimensional import MMM

# the Bayesian MMM: adstock, then saturation
mmm = MMM(date_column="week", target_column="sales",
          channel_columns=["google", "meta"],
          control_columns=["week_no"],
          adstock=GeometricAdstock(l_max=6),
          saturation=LogisticSaturation(),
          yearly_seasonality=2)
X = at.select("week", "google", "meta", "week_no")
mmm.fit(X.to_pandas(), at["sales"].to_pandas(),
        draws=500, tune=500, chains=2,
        random_seed=1, progressbar=False)
```

setting	value
chains × draws	2 × 500
adstock window	6 weeks
season	yearly, 2 waves
runtime	about 25 s

One model, all at once: adstock, saturation, trend and season, with priors.

Step 9: check and read

open_source_mmm.qmd · Step 9

```
import arviz as az

# the decay of each channel, with R-hat
print(az.summary(mmm.idata,
                 var_names=["adstock_alpha"]))

# ROI per channel: mean and 94 % interval
roi = mmm.summary.roas(frequency="all_time")
roi
```

result	google	meta
decay	0.05	0.11
R-hat	1.00	1.00
ROI	3.38	9.45
94 % interval	2.33–4.65	6.66–12.18

Same ranking as OLS, now with a range: and Meta's decay is 0.11, not 0.5.

In-class assignment

Quiz, four tasks, no coding

Online quiz: MMM in five questions

1 Decay 0.5, last week's adstock 6, spend this week 10. This week's adstock is …
A 16 B 13 C 8

2 A channel brought 120 k EUR of sales for 40 k EUR of spend. Its ROI is …
A 0.33 B 3.00 C 80

3 Sales 30 at 10 k EUR a week, 36 at 20 k EUR a week. The ROI of the extra 10 k EUR is …
A 3.00 B 1.80 C 0.60

4 Which tool fits many ridge regressions and lets the analyst pick from a Pareto front?
A PyMC-Marketing B Meridian C Robyn

5 A new market has six months of weekly data. The best start is …
A a model of its own B partial pooling with the other markets C the biggest market's ROI

3 minutes alone, then compare with your neighbour.

Answers: in class

Your turn: an MMM review in four tasks

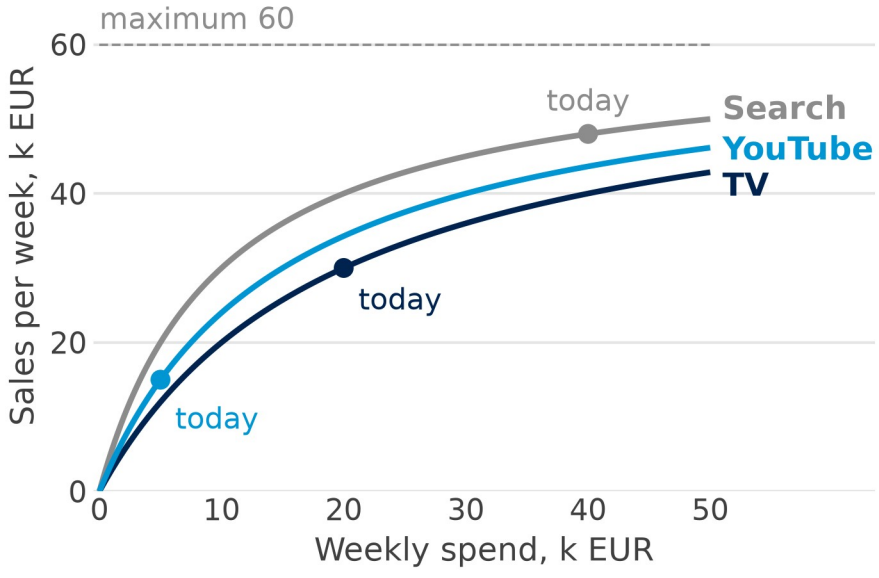
Groups of three: 35 minutes	Plenary: 10 minutes

1. **Budget:** three response curves of an e-bike brand: where should 10 k EUR a week go? (slide 44)
2. **Model check:** the German model credits Google and Meta with 95 % of sales. Plausible? What would you test?
3. **Vendor claim:** grade one case study with the four questions (slide 24)
4. **Tool choice:** PyMC-Marketing, Meridian or Robyn for two brands, with two reasons each (slide 23)

Deliverable: one slide per group: the budget move, the verdict on Germany, the grade, the two tools.

Handout: `sessions/02-mmm-one-country/mmm_assignment.qmd` (no code needed)

Task 1: three response curves



Now	Spend	Sales	Avg ROI
TV	20	30	1.50
YouTube	5	15	3.00
Search	40	48	1.20

Sales at	−10 k EUR	+10 k EUR
TV	20	36
YouTube	–	30
Search	45	50

Your move: which channel gives 10 k EUR a week, which one gets it, and why?

Invented round numbers (assignment)

Open data for MMM projects

Conjura eCommerce MMM

Real, anonymised daily data of about 100 web shops; Google, Meta, TikTok

Licence: CC BY 4.0 (our real data)

Meridian geo data

Simulated weekly panels, many regions, reach and frequency

Use: geo vs national model

PyMC-Marketing data

A two-region panel with true parameters, lift-test data

Use: can the model find the truth?

Robyn demo data

Simulated weekly spend and revenue; holidays of 123 countries

Use: an R lab, holiday effects

The licence decides: commit only data you may share; link the rest.

Country controls: Eurostat and World Bank (prices, population, internet use)

Sources: dataset pages; links in open_source_mmm.qmd (appendix)

Key takeaways

1. **One line:** sales = baseline + media effects + controls + noise
2. **Carryover and saturation:** spend keeps working, and each extra euro buys less
3. **Decide at the margin:** the next euro, not the average ROI
4. **Three tools:** two Bayesian tools that pool markets, one ridge tool per market
5. **Be sceptical:** spend follows demand; calibrate with experiments; ask who wrote the case

Next: run sessions/02-mmm-one-country/open_source_mmm.qmd; it reproduces Parts 4 and 5.

References

MMM methods

Jin, Y., Wang, Y., Sun, Y., Chan, D. & Koehler, J. (2017). Bayesian methods for media mix modeling with carryover and shape effects. Google Inc.

Sun, Y., Wang, Y., Jin, Y., Chan, D. & Koehler, J. (2017). Geo-level Bayesian hierarchical media mix modeling. Google Inc.

Zhang et al. (2024). Google Research: calibration with Bayesian priors (ROI priors from lift tests).

Runge, Skokan, Zhou & Pauwels (2024). arXiv 2403.14674: the method behind Robyn.

Runge & Pauwels (2026). Customer Needs and Solutions; free as MSI Working Paper 26-106.

Gelman et al. (2020). Bayesian Workflow, arXiv 2011.01808.

Tools and data

PyMC-Marketing (PyMC Labs). Apache-2.0, github.com/pymc-labs/pymc-marketing.

Meridian (Google). Apache-2.0, github.com/google/meridian.

Robyn (Meta Marketing Science). MIT, github.com/facebookexperimental/Robyn.

Anderson, A. (2024). Multi-Region Marketing Mix Modelling (MMM) Dataset for Several eCommerce Brands (v3). figshare. doi.org/10.6084/m9.figshare.25314841.v3. CC BY 4.0.

Free to read: MSI Working Paper 26-106 (Runge & Pauwels), arXiv 2011.01808 (Bayesian workflow)

Thank you. Questions and discussion

VIENNA UNIVERSITY OF ECONOMICS AND BUSINESS

INSTITUTE FOR INTERNATIONAL BUSINESS
Welthandelsplatz 1, 1020 Vienna, Austria

DR ARNE FLOH
Senior Lecturer in Marketing

F +43-31336-6367
arne.floh@wu.ac.at
www.wu.ac.at/en/welthandel/arne-floh
Office hours: book online